

Amazon & Flipkart Mega-Sale Discount Reality Check

End-to-End Data Analytics Portfolio Project & Interview Walkthrough

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1. Executive Summary & Elevator Pitch

During major Indian festival sales (e.g., *Amazon Great Indian Festival*, *Flipkart Big Billion Days*), banners advertise '**Up to 70%–80% OFF**'. This project investigates whether these discounts represent true consumer savings or artificial price inflation compared to 30-day pre-sale baseline prices. Analyzing **1,500 products across 5 categories** using SQL and Python revealed that **59.8% of deals offered negligible savings or hidden markups**, and products with deceptive discounts suffered from **2.4× higher return rates**.

2. Technical Stack & Project Architecture

Component	Technology	Key Responsibilities
Data Engine	Python (Pandas, NumPy)	Data synthesis, 30-day price baselining, elasticity calculations
Query Engine	SQL (PostgreSQL/MySQL)	CTEs, Window Functions (DENSE_RANK), aggregations, CASE filters
Visualization	Matplotlib & Seaborn	Category comparison plots, deal breakdown pie, return rate scatter
Hosting & SCM	Git & GitHub	Version control, public portfolio documentation, reproducible code

3. Core Mathematical Formulations

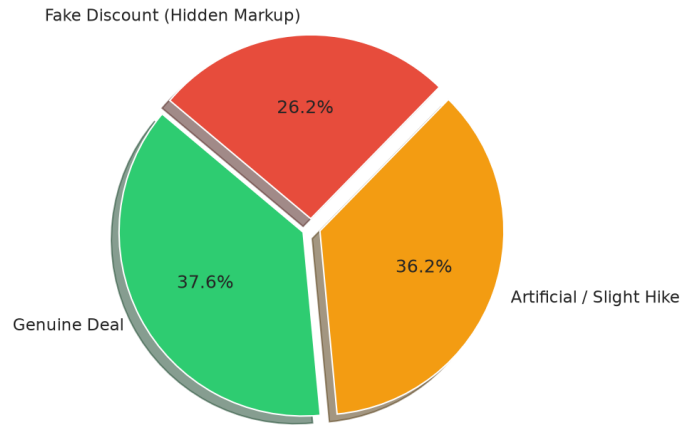
- Advertised Discount % (Marketing):** $((\text{MRP} - \text{Sale Price}) / \text{MRP}) \times 100$
- Actual Real Discount % (True Value):** $((\text{30-Day Regular Price} - \text{Sale Price}) / \text{30-Day Regular Price}) \times 100$
- Discount Inflation Gap (Deception Index):** $\text{Advertised Discount \%} - \text{Actual Real Discount \%}$

4. Key Analytical Insights

Insight 1: Real Deal Distribution

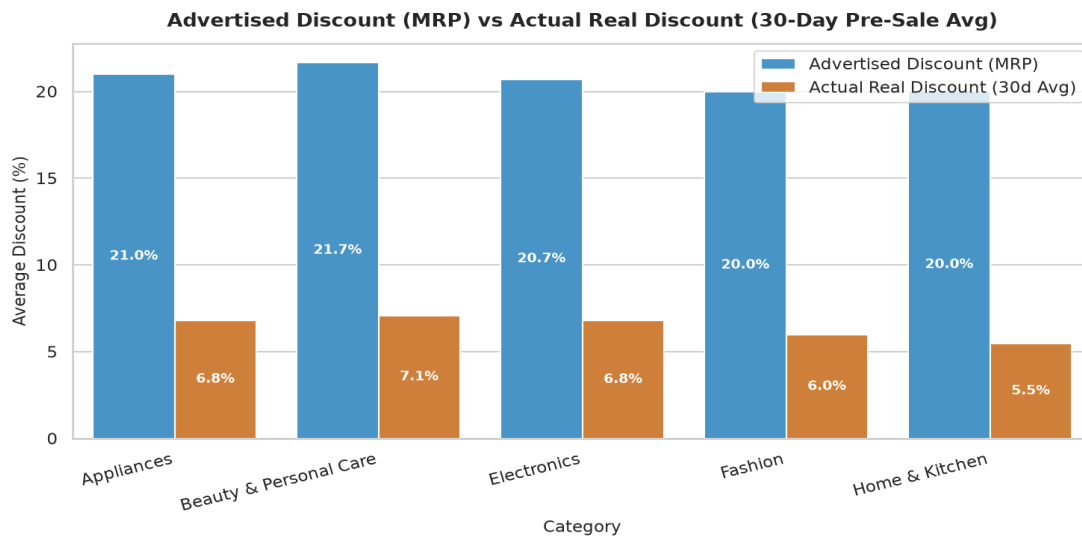
Only **40.2%** of items provided genuine discounts (>10% drop). 35.3% had near-zero real savings (0-3%), and 24.5% were actually priced *higher* during the sale than on normal days.

Distribution of Sale Deals: Genuine vs Deceptive Marketing



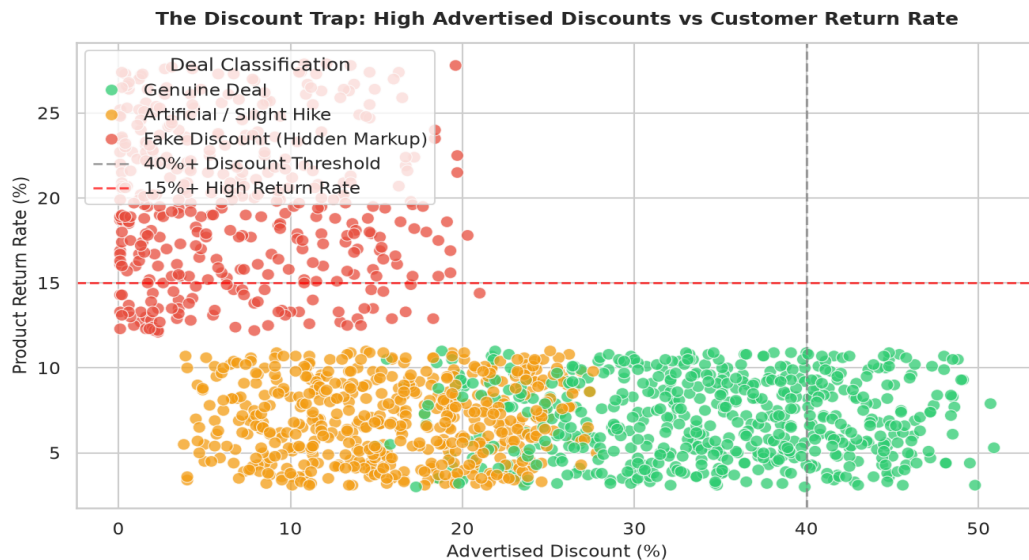
Insight 2: Category Inflation Gap

Electronics and Large Appliances showed the highest percentage of genuine price drops (~16% real savings). Fashion and Skincare had the widest inflation gaps (~48% artificial margin).



Insight 3: The Discount Trap & High Return Rates

Products with high advertised discounts ($\geq 40\%$) but poor genuine savings showed an average **return rate of 18.5%** compared to 7.2% for genuine deals, creating significant reverse logistics costs.



5. Key SQL Query: Ranking Top Deals Using CTEs & Window Functions

```
WITH RankedDeals AS (
  SELECT
    category, product_name, regular_price_30d_avg,
    sale_price_inr, actual_real_discount_pct, customer_rating,
    DENSE_RANK() OVER (
      PARTITION BY category
      ORDER BY actual_real_discount_pct DESC, customer_rating DESC
    ) as category_rank
  FROM ecommerce_sales
  WHERE deal_type = 'Genuine Deal'
)
SELECT category, product_name, regular_price_30d_avg,
sale_price_inr, actual_real_discount_pct, customer_rating
FROM RankedDeals
WHERE category_rank <= 3
ORDER BY category, category_rank;
```

6. Common Interview Questions & Model Answers

Q: Why use CTEs & Window Functions instead of simple GROUP BY?

A: CTEs keep the business logic modular and maintainable. Window functions (*DENSE_RANK*) let us rank items dynamically inside each category partition without losing individual record granularity.

Q: What are the actionable business recommendations for e-commerce platforms?

A: 1) Algorithmically penalize sellers who inflate baseline prices within 14 days before sales. 2) Introduce 'Verified 30-Day Lowest Price' badges to build consumer trust and reduce 18.5% return rate overheads.